Suggested Changes Before Submission

Internal checklist — 18 October 2026

The following checklist reflects the current draft. Resolved requests have been removed, and partially completed items now identify only the remaining changes. The current main document compiles successfully. This checklist does not implement changes to the paper.

1. SIGMOD Paper Style Issues

1. **Meet the page limit.** The current draft's conclusion extends onto page 13. Reduce the main paper to 12 pages excluding references, as required by the SIGMOD call. Shorten repeated discussion and remove redundant material; do not alter the ACM template's fonts or spacing.
2. **Inspect the final submission files.** After the remaining edits and appendix separation, check anonymity and page transitions in the exported PDFs. Upload the final version and keep this internal checklist out of the submission.

2. Experimental Section Issues

3. **Finish the polynomial experiment or remove its empirical claim.** The linear optimum-comparison table is now present, but the polynomial subsection in `sections/experiment.tex` still contains a red placeholder. Add verified results, including the degree and dataset used, or remove the pending subsection and the experimental summary's claim that small polynomial experiments demonstrate effectiveness. The polynomial appendix currently describes methods without numerical results. Finalize the other red experimental passages before submission.
4. **Replace the empty code link.** Provide an anonymous artifact URL, or remove the empty footnote and explain artifact availability in the submission form.
5. **Correct the synthetic optimum reference.** An expected 5% inversion count is not an instance-wise upper bound. Report the generating score's realized distance, or rename the row *5% expected inversion reference* and remove the upper-bound claim from the synthetic table's caption. In the dataset appendix, also replace the claim that the selected permutation has 5% inversions with the corresponding expectation statement.
6. **Clarify the new exact-optimum comparison.** In the paragraph accompanying `tab:OPT-linear`, distinguish the two objective ratios. With $n = 100$ and optimum disagreement 286, the optimum agreement is $4950 - 286 = 4664$. CLR's disagreement ratio is $293/286 \approx 1.0245$; Approx-Agree's agreement ratio is $4659/4664 \approx 0.9989$, which exceeds the $1 - \varepsilon = 0.75$ target, rather than a maximization guarantee of 1.25. The half-approximation's agreement ratio is $(4950 - 2143)/4664 \approx 0.6018$. State the ILP solver's optimality status, runtime, and effective dimension; describe these as observations on this instance, not an empirical proof of the worst-case guarantees. Separate OPT and Local Search in the first row, define CLR, and use *Kendall's Tau* in the column heading.
7. **Address the closest prior method.** Compare with RANKHOW on a compatible configuration, or explain precisely why that comparison is unavailable. Generic learning-to-rank baselines do not replace this comparison.
8. **Report an actual quality–time trade-off.** Add end-to-end runtimes or quality at common time budgets, including pilot computation and full-data evaluation. Maximum sample size under

a timeout alone does not establish faster execution.

9. **Revise the remaining performance overclaims.** The revised real-world discussion correctly notes dataset-dependent winners. Align the remaining overall claims and the experimental summary with that qualification. In particular, replace the claimed algorithm-independent *monotone property* with an observed tendency for larger samples to improve quality. The red additive-bonus discussion still attributes stability and variation to convergence and iteration limits without isolating those effects. Interpret improvements alongside their size and reported variation, and explain that the half-approximation guarantees agreement rather than a minimum-disagreement ratio. Also complete the unfinished ablation sentence ending *account for some of the .*

10. **Check identical rows and seed variation.** Verify the matching synthetic results for Sampling, Weighted Sampling, and LMRC, and for RankNet and RankSVM. Explain confirmed causes. For the half-approximation, state that changing the seed selects a different valid starting direction.

11. **Add the remaining reproducibility details.** Give the actual sample-size schedule, feature scaling, categorical encoding, regularization, baseline stopping rules, solver tolerances, software versions, and thread counts. Distinguish data seeds from algorithm seeds and describe efficient full-data inversion counting. Cross-reference the class-multiplicity and proposal-weight formula already given in the polynomial appendix. The timeout, five seed values, and weighted-search restart and iteration limits are already specified.

12. **Clarify dataset construction.** Document Retiring Adult's exact task and attribute-count convention; identify which NBA target the table evaluates; specify JEE bonus formulas and whether group attributes enter learned scores; document HCP's provenance. Keep the reported Retiring Adult counts unchanged, as requested; clarify their meaning rather than guessing missing facts.

3. Theory, Introduction, and Preliminaries Issues

13. **Qualify the half-agreement claim for duplicate rows.**
Where to modify: In `sections/experiment.tex`, revise the last sentence of the circle-sweep description and state the counting convention in Experimental Setup. The theoretical scope clarification in `sections/prelims.tex` is already in place; no replacement is needed.
Suggested clarification: Let M count target-strict pairs, U count those with identical feature vectors, and $P = M - U$. Comparing a score with its reverse guarantees at least $P/2$ agreements, not necessarily $M/2$, because the U duplicate pairs remain forced disagreements. This still gives half of the optimal agreement, since the optimum is at most P. Keep the distinct-vector theoretical statement unchanged and follow the distinction already made in the polynomial-experiment appendix.

4. Appendix Issues

14. **Export the appendix separately.** Keep only the main paper and references in the main submission PDF. Submit a focused appendix separately and preserve its cross-references. The main paper must remain understandable without it.

15. **Correct references to the appendix's result tables.** In `sections/experiment.tex`, the later HCP and NBA summaries use *Appendix Table* but reference `subsubsec:append-unknown-scoring-models`,

a subsection label. Use `tab:kt-method-comparison-hcp` and `tab:kt-method-comparison-nba`, respectively, as the earlier paragraph already does.

16. **Synchronize the experimental appendix with the main text.** In `sections/appendix-exps.tex`, replace the obsolete Hybrid name with Sampling-CLR. Explain Approximate-Agreement's limitations using the actual sample budget and optimizer; the current attribution to *smaller values of* ε does not clearly explain its poorer observed quality.

Scope: Updated against the current manuscript, included appendix material, compilation, and reference checks. The small-instance ratios above are calculated from the reported table; the benchmarks and claimed exact optimum were not independently reproduced or certified.